Adnan Harun Doğan

Personal Information

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Employment

Feb 2023–Present — **Research Assistant**, Image Processing and Pattern Recognition Laboratory (ImageLab), METU
Supervisor: Prof. Dr. Sinan Kalkan, Assoc. Prof. Dr. Emre Akbas — *Ankara, Turkey*

Jan 2025–Oct 2025 — **Research Intern**, Sensing, Interaction, & Perception Laboratory (SIPLab), ETH Zürich
Supervisor: Prof. Dr. Christian Holz — *Zürich, Switzerland*

Oct 2021–Feb 2022 — **Intern**, Center for Artificial Intelligence and Robotics (ROMER), METU
Supervisor: Assoc. Prof. Dr. Erol Sahin, Assoc. Prof. Dr. Bugra Koku — *Ankara, Turkey*

Jun 2021–Sep 2021 — **Research Intern**, Cyber-Physical Systems Laboratory (AICPS), University of California, Irvine
Supervisor: Prof. Dr. Mohammad Abdullah Al-Faruque — *Irvine, California, USA*

Education

2022–2025 — **Master of Science in Computer Engineering**, Middle East Technical University (METU)
Advisor: Prof. Dr. Sinan Kalkan, Co-advisor: Assoc. Prof. Dr. Emre Akbas — *Ankara, Turkey*
Thesis: Incorporating Piecewise Linear Functions with Constant Regions in Backpropagation

2018–2022 — **Bachelor of Science in Computer Engineering**, Middle East Technical University (METU)
Senior Project: ReQuest: Automated Question-Answer Generation from Text — *Ankara, Turkey*

Selected Graduate Coursework

Optimization: Linear Optimization (IE 553), Numerical Optimization (IAM 566), Optimization Methods for ML (IAM 771)
Probability and Statistics: Probability Theory (IAM 541), Decision Theory and Bayesian Analysis (STAT 565)
Deep Learning: Deep Learning (CENG 501), Advanced Deep Learning (CENG 502), Deep Generative Models (CENG 796)

Research Interests

My research has focused on gradient-based learning through non-differentiable operations. I am interested in problems where the model or its loss contains a discrete decision (e.g., ordering detections by confidence, assigning predictions to targets, choosing which layers a token passes through, or deciding where to refine a mesh). Such decisions give backpropagation little or no gradient signal. I have designed structure-respecting surrogate gradients for the building blocks behind them, such as ranking, sorting, matching and routing, and studied how this choice shapes what the network learns. Continuous relaxations of these decisions also lead me to optimal transport and measure-theoretic tools.

Keywords: gradients through discrete decisions, differentiable optimization, decision-focused learning, optimal transport, neural state estimation, differentiable ranking and matching, conditional computation

Publications

Selected publications

7. **Dogan, A. H.**, Demirel, B. U., & Holz, C. (2026). *Learned-noise Kalman filtering with a body prior for drift-calibrated full-body pose from raw sparse inertial signals.* Submitted[†].
6. **Dogan, A. H.**, Arasli, D., Oksuz, K., Akbas, E., & Kalkan, S. (2026). *Variational surrogate Jacobians: A framework for constrained gradient propagation.* Submitted[†].
5. **Dogan, A. H.**, Deniz, M., Alemdar, H., & Baran, O. U. (2026). *CoFINN: Conservation flux informed neural networks for physics problems governed by conservation laws.* **arXiv preprint**. arXiv:2607.06587
4. **Dogan, A. H.**, Demirel, B. U., & Holz, C. (2026). Frequency-weighted neural Kalman filters. ***International Conference on Robotics and Automation (IEEE ICRA)***. arXiv:2606.02251 (PDF, Code)
3. **Dogan, A. H.**, Celtik, S. O., Baran, O. U., & Alemdar, H. (2026). *Prediction-driven adaptive mesh refinement with graph neural networks for transonic airfoil flows.* Submitted to Journal of Aerospace Science and Technology.
2. Demirel, B. U., **Dogan, A. H.**, Rossie, J., Möbus, M., & Holz, C. (2025). Beyond subjectivity: Continuous cybersickness detection using EEG-based multitaper spectrum estimation. ***IEEE Transactions on Visualization and Computer Graphics (IEEE TVCG), 31***(5), 2525-2534. doi.org/10.1109/TVCG.2025.3549132 (arXiv, Code)
1. Yavuz, F., Cam, B. C., **Dogan, A. H.**, Oksuz, K., Kalkan, S., & Akbas, E. (2024). Bucketed ranking-based losses for efficient training of object detectors. ***European Conference on Computer Vision (ECCV)***, Lecture Notes in Computer Science, vol 15117, pp. 93-109. doi.org/10.1007/978-3-031-73202-7_6 (arXiv, PDF, Code)

Other publications

6. Amin, M. A., **Dogan, A. H.**, Kuru, E. S., Sever, Y., & Angin, P. (2024). Misuse detection and response for orchestrated microservices based software. *International Conference on Advanced Information Networking and Applications (AINA)*, Lecture Notes on Data Engineering and Communications Technologies, vol 204, pp. 217-226. doi.org/10.1007/978-3-031-57942-4_22 (Data)
5. Karagoz, P., Cekinel, R. F., **Dogan, A. H.**, Oktay, B., Ozturk, A. U., Tonay, T. S., & Tuncel, B. M. (2023). Enhancing underground built heritage analysis with text mining: A case study on Cappadocia. In M. Akkar Ercan & K. Aydiner (Eds.), *Valorising Underground Built Heritage in Cappadocia* (pp. 61-92). CNR Edizioni. ISBN 978-88-8080-605-9. (PDF)
4. Sever, Y., & **Dogan, A. H.** (2023). A Kubernetes dataset for misuse detection. *ITU Journal on Future and Evolving Technologies, 4*(2), 383-388. doi.org/10.52953/fplr8631 (PDF, Code)
3. Sever, Y., Ekinci, G., **Dogan, A. H.**, Alparslan, B., Gurbuz, A. S., Jabrayilov, V., & Angin, P. (2022). An empirical analysis of IDS approaches in container security. *International Workshop on Secure and Reliable Microservices and Containers (SRMC)*, pp. 18-26. doi.org/10.1109/SRMC57347.2022.00007 Best Paper Award
2. Demirel, B. U., **Dogan, A. H.**, & Al-Faruque, M. A. (2021). Two might do: A beat-by-beat classification of cardiac abnormalities using deep learning and domain-specific features. *Computing in Cardiology (CinC)*, pp. 1-4. doi.org/10.23919/cinc53138.2021.9662935 (PDF, Code)
1. Buyukbas, E. B., **Dogan, A. H.**, Ozturk, A. U., & Karagoz, P. (2021). Explainability in irony detection. *Big Data Analytics and Knowledge Discovery (DaWaK)*, Lecture Notes in Computer Science, vol 12925, pp. 152-157. doi.org/10.1007/978-3-030-86534-4_14

Theses

1. **Dogan, A. H.** (2025). *Incorporating piecewise linear functions with constant regions in backpropagation.* **M.Sc. Thesis, Middle East Technical University**. (Thesis)

† Venue withheld while under double-blind review.

Undergraduate Research Projects

Sep 2021–Jun 2022

High Performance Intrusion Detection and Prevention for Software-Defined Container Networks in the Cloud, METU

CONTSEC (TUBITAK 3501): security of containerized microservices on Kubernetes

- Designed and implemented an intrusion detection pipeline for containerized cloud systems
- Built a public Kubernetes misuse dataset of network flows from 10 labelled attack scenarios

Advisor: Assoc. Prof. Dr. Pelin Angin — *Ankara, Turkey*

Feb 2021–Jun 2022

Image Processing and Pattern Recognition Laboratory (ImageLab), METU

Conservation flux informed neural networks and adaptive mesh refinement for transonic airfoils

- Designed an HLLC Riemann solver momentum loss on pixels, cutting drag error by 15%
- Trained CNN, FNO, ViT and diffusion models with and without the momentum flux loss
- Refined GNN airfoil meshes at Ducros-flagged shocks, cutting the area-based RMSE by 12%

Advisor: Assist. Prof. Dr. Hande Alemdar, Assist. Prof. Dr. Ozgur Ugras Baran — *Ankara, Turkey*

Oct 2020–Jun 2021

Textual Event Graph HUB (TEGHub), METU

Textual analysis for irony detection and research impact prediction

- Adapted T5 to Turkish irony detection, its first use for the task, reaching 59.5% accuracy
- Fine-tuned BERT with Simple Transformers, reaching 69.0% accuracy on Turkish irony
- Crawled scientific papers and their Twitter mentions as data for research impact prediction
- Applied LDA, K-Means and rule-based VADER sentiment to 1,742 Cappadocia reviews

Advisor: Prof. Dr. Pinar Karagoz — *Ankara, Turkey*

Feb 2020–Jun 2021

School of Medicine, Koç University

Healthcare policy simulation for evidence-based decision making

- Used simulation models focusing on mechatronics applications in the healthcare domain
- Built the citation graph of 999 health systems papers from Microsoft Academic Graph data
- Developed GCN-based node embeddings on this citation graph of health systems literature

Advisor: Assist. Prof. Dr. Ozge Karanfil — *Istanbul, Turkey*

Teaching

Teaching Assistant, Computer Engineering Department, METU Spring 2023–Present

8. CENG 424 - Logic for Computer Science (Fall 2023, Fall 2025, Fall 2026)
7. CENG 280 - Automata Theory (Spring 2024, Spring 2025, Spring 2026)
6. CENG 242 - Programming Language Concepts (Spring 2023, Spring 2025, Spring 2026)
5. CENG 223 - Discrete Computational Structures (Fall 2023, Fall 2024, Fall 2025)
4. CENG 460 - Introduction to Robotics for Computer Science (Fall 2024)
3. CENG 213 - Data Structures (Spring 2024)
2. CENG 382 - Analysis of Dynamical Systems (Spring 2023)
1. CENG 334 - Introduction to Operating Systems (Spring 2023)

Undergraduate Teaching Assistant (Volunteer), Computer Engineering Department, METU Spring 2019–Fall 2021

3. CENG 240 - Programming with Python for Engineers (Fall 2021)

2. CENG 111 - Introduction to Computer Engineering Concepts (Fall 2020)
1. CENG 230 - Introduction to C Programming (Spring 2019)

Honors and Scholarships

2022–2023	TUBITAK 2247-C STAR'23 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2021–2022	TUBITAK 2247-C STAR'22 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2020–2021	TUBITAK 2247-C STAR'21 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2017	University Entrance Exam: ranked 1466th of 2.27M candidates by OSYM, Turkey

Computer Skills

Languages:	Python, C/C++, Rust
Tools & Libraries:	PyTorch, CVXPYLayers, Gurobi, OR-Tools, POT, MNE, ROS2
DevOps & HPC:	GitHub, Docker, Slurm (Workload Manager), Apptainer

Languages

Turkish:	Native
English:	Fluent (TOEFL iBT 109/120, 2022)